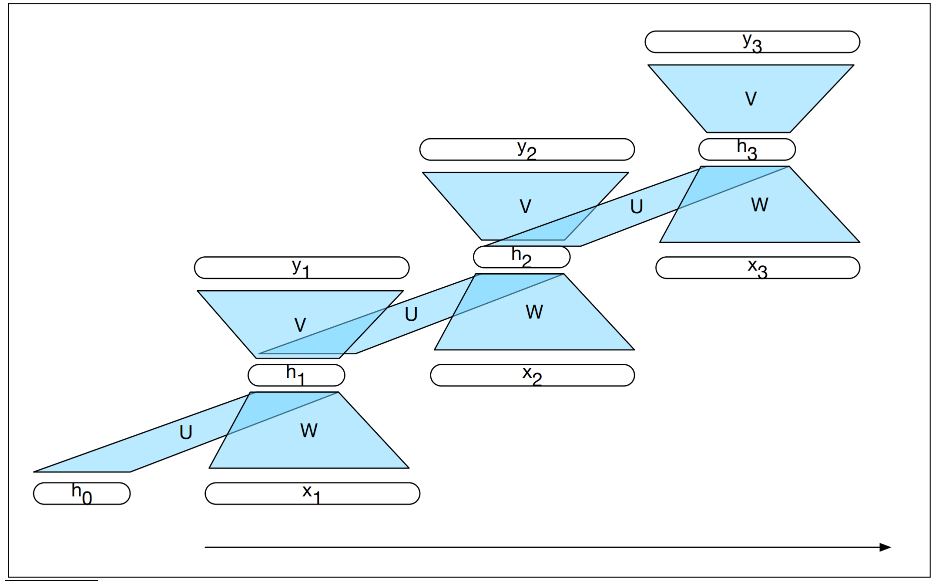
A word on Backpropogation Through Time

- In order to train the network, we need to decide what output of the network we would like to compare to the gold standard and what loss function to use
- Even if the recurrent function R and the output function O are shallow (e.g., a single affine function), the computation graph may be deep due to the recurrence
- It is common practice to limit (truncate) the number of iterations
- We will see an example later



Reminder: Loss Functions

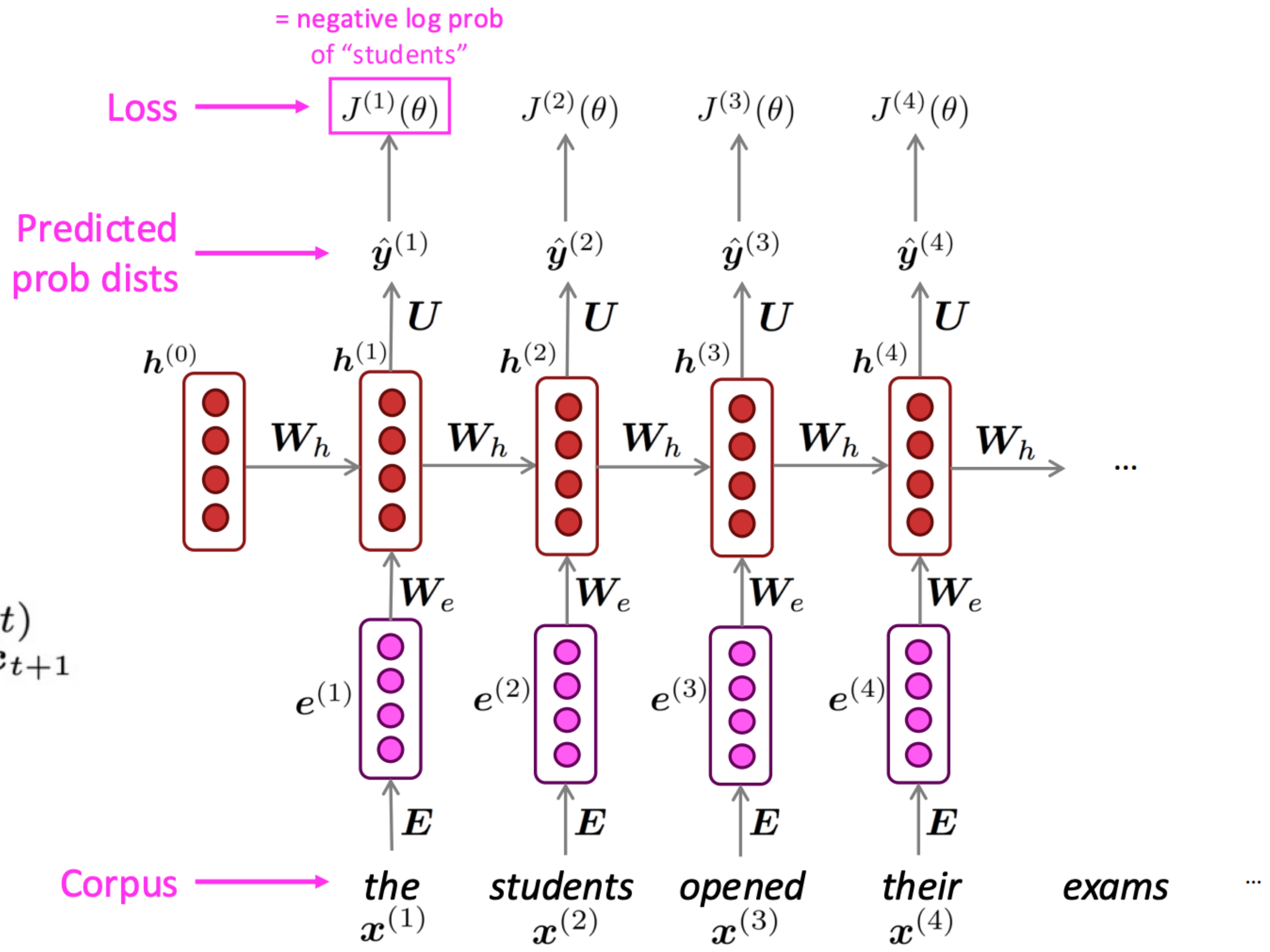
- Training is done by defining a loss function (i.e. a function that determines how “bad” a classification is relative to the gold standard)
- The network then attempts to minimize the empirical risk. For a sample $S=\{x_1,\ldots,x_m\}$ with gold standard labels $\{y_1,\ldots,y_m\}$, the empirical risk is:

$$\frac{1}{m}\sum_{i=1}^{m}\ell(\theta;(x_i,y_i))$$

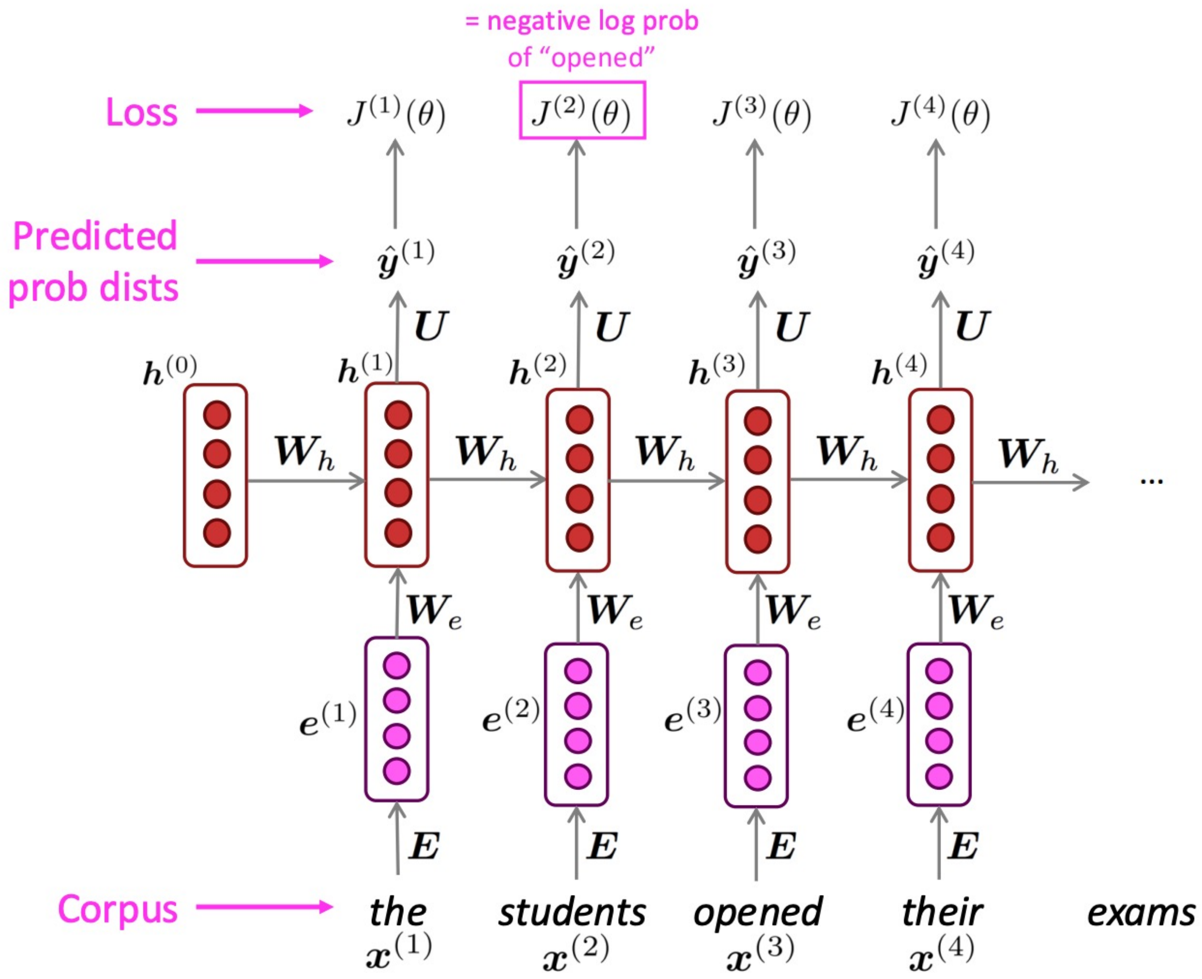
Training RNN LMs

- Neural language models are generally trained to maximize the sum of the predicted log-probabilities of the correct words
  - Cross-entropy loss

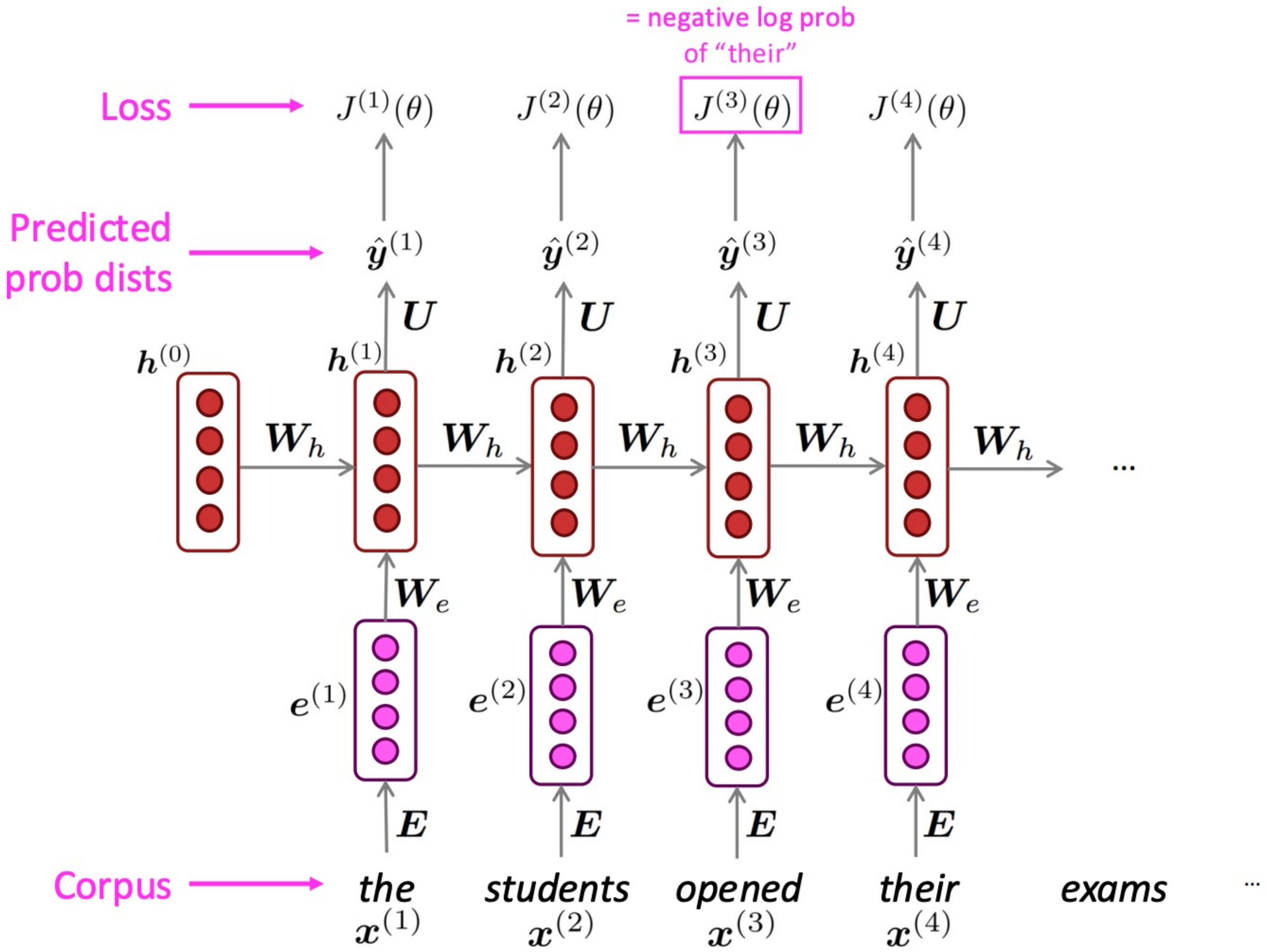
$$J(\theta) = \frac{1}{T}\sum_{t=1}^{T} J^{(t)}(\theta) = \frac{1}{T}\sum_{t=1}^{T} -\log \hat{\boldsymbol{y}}^{(t)}_{\boldsymbol{x}_{t+1}}$$



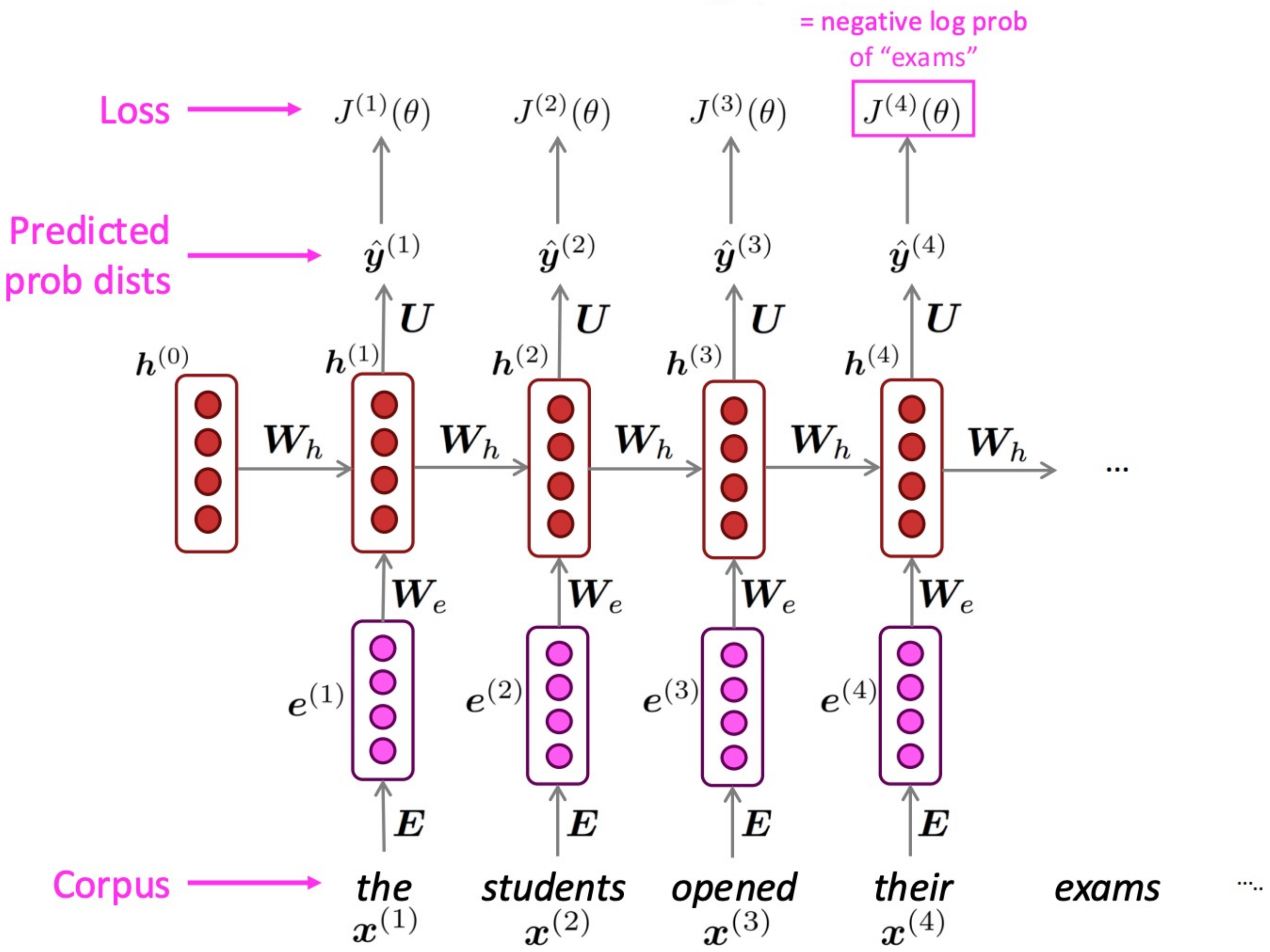
Training RNN LMs



Training RNN LMs



Training RNN LMs



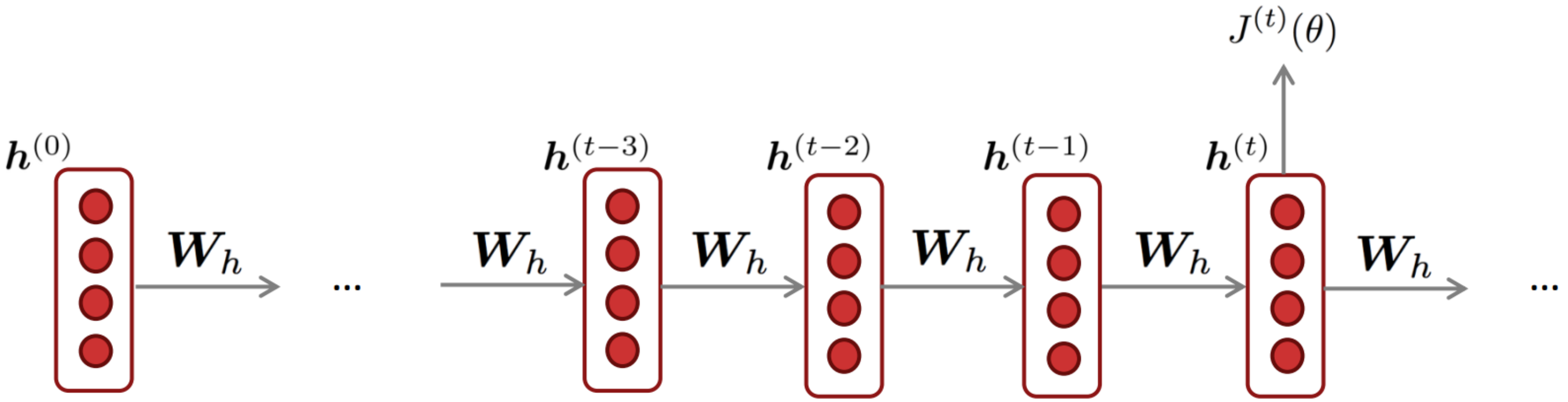
Training RNN Language Model

- However:
  - Computing loss and gradients across entire corpus at once is too expensive (memory-wise)!

$$J(\theta) = \frac{1}{T} \sum_{t=1}^{T} J^{(t)}(\theta)$$

- In practice, consider as a sentence (or a document)

Backpropagation for RNNs

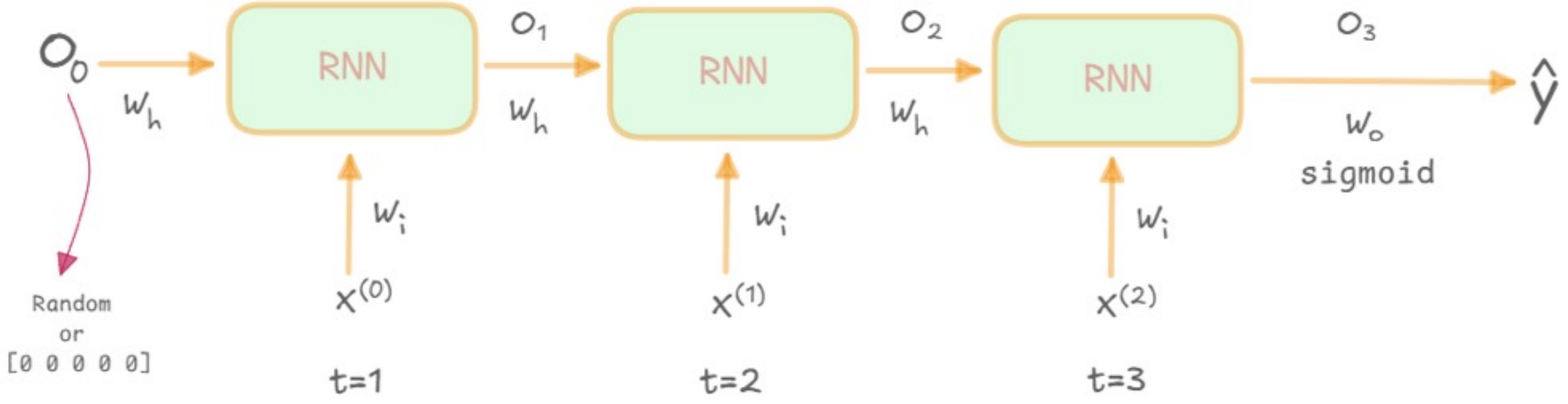


<u>Question:</u> What's the derivative of $J^{(t)}(\theta)$ w.r.t. the repeated weight matrix $\boldsymbol{W}_h$?

<u>Answer:</u> $$\frac{\partial J^{(t)}}{\partial \boldsymbol{W_h}} = \sum_{i=1}^{t} \left. \frac{\partial J^{(t)}}{\partial \boldsymbol{W_h}} \right|_{(i)}$$

"The gradient w.r.t. a repeated weight
is the sum of the gradient
w.r.t. each time it appears"

Backpropogation for RNNs: Simple Case



$$O_1 = f\left(X_i^{(0)} W_i + O_0 W_h\right)$$

$$O_2 = f\left(X_i^{(1)} W_i + O_1 W_h\right)$$

$$O_3 = f\left(X_i^{(2)} W_i + O_2 W_h\right)$$

$$\hat{y} = \sigma\left(O_3 W_0\right)$$

https://medium.com/@imsanketsingh/day-5-backpropagation-through-time-bptt-0cad57696ecd

Backpropogation for RNNs: Simple Case

- Assume *L* is the loss

$$\frac{d\mathcal{L}}{dW_h} = \frac{d\mathcal{L}}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_3} \cdot \frac{dO_3}{dW_h}$$
$$+ \frac{d\mathcal{L}}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_3} \cdot \frac{dO_3}{dO_2} \cdot \frac{dO_2}{dW_h}$$
$$+ \frac{d\mathcal{L}}{d\hat{y}} \cdot \frac{d\hat{y}}{dO_3} \cdot \frac{dO_3}{dO_2} \cdot \frac{dO_2}{dO_1} \cdot \frac{dO_1}{dW_h}$$